RV UNIVERSITY

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

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CONTINUOUS INTERNAL EVALUATION – 2 (CIE – 2)

MERCEDES SILVER ARROWS

Interactive 3D & Photographic Luxury Vehicle Showcase and Studio Configurator Platform

Course: Full Stack Development (FSD) • Academic Year 2026–2027

CANDIDATE INFORMATION

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Program:	Bachelor of Computer Applications
Semester:	III Semester (2024–2028 Batch)
Specialization:	Full Stack Web Engineering

EVALUATION & COURSE DETAILS

Faculty Guide:	Mohammed Danish
Designation:	Course Instructor / Faculty
Department:	Dept. of Computer Applications
Submission:	Soft Copy Report in PDF
Date:	October 12, 2026

1. PROJECT OVERVIEW & PROBLEM STATEMENT

1.1 Executive Summary

Mercedes Silver Arrows is an advanced, enterprise-grade luxury automotive digital showroom and interactive vehicle configurator built with modern full-stack web technologies. Conceived and implemented to mirror and exceed the high visual standards of the official *Mercedes-Benz USA (MBUSA)* digital experience, the platform enables potential buyers and automotive enthusiasts to inspect, filter, customize, and price luxury vehicles across all major segments — including all-electric EQ innovators, high-performance AMG vehicles, executive sedans, rugged SUVs, and ultra-luxury Maybach models.

The web application integrates high-resolution studio photography with a responsive multi-angle rendering pipeline, realistic metallic reflection overlays, live MSRP and monthly financing calculations, interactive 360-degree driver cockpit inspection, and state persistence using modern browser technologies.

1.2 Problem Statement & Motivation

Modern consumer expectations for digital automotive commerce demand photorealistic visual fidelity, instantaneous reactivity, and seamless mobile accessibility. However, conventional 3D automotive configurators in the industry suffer from several critical shortcomings:

- **Excessive Bandwidth & Heavy 3D Assets:** Downloading multi-gigabyte WebGL models results in 8–15 second initial load times and severe thermal throttling on mobile devices.
- **Flicker & Visual Glitches During Customization:** Unoptimized paint and wheel changes frequently cause white flashes or broken texture swaps, degrading perceived brand luxury.
- **Lack of Real-Time Financial Feedback:** Many configurator platforms disconnect option selection from realistic transparent pricing, requiring users to navigate away to estimate payments.
- **State Loss & Poor Bookmarking:** Users unable to bookmark specific trim-level builds or save configurations locally lose their customized selections upon page reloads.

1.3 Project Objectives

🎯 Seamless Catalog Exploration

Deliver instantaneous category filtering across 24 vehicle families and 77 distinct trims with live powertrain indicators (All-Electric, Hybrid, Biturbo Gas).

🎨 Sub-100ms Studio Customization

Enable fluid, flicker-free switching between authentic manufacturer finishes (Metallic, Gloss, and MANUFAKTUR Magno Matte) via intelligent image preloading.

🔄 Interactive Multi-Angle Views

Offer 3/4 front exterior, side profile with custom wheel fitments, rear view, and a 60fps inertia-damped panoramic driver cockpit interior.

💾 Full State Synchronization & Storage

Synchronize URL query parameters (`?build=`, `?trim=`) and persist completed configurations into client-side `localStorage`.

2. SYSTEM ARCHITECTURE & TECHNOLOGY STACK

2.1 Technology Stack Architecture

TIER / DOMAIN	TECHNOLOGY	VERSION	ROLE & ENGINEERING SIGNIFICANCE
Frontend Core	React	19.2.8	Declarative component-based UI with concurrent rendering and optimized Hooks.
Build Tooling	Vite	8.3.0	Next-gen frontend tooling providing sub-200ms dev HMR and 680ms production builds.
3D Graphics	Three.js & R3F	0.186 / 9.8	Hardware-accelerated WebGL rendering for 3D stage elements and star animations.
Code Quality	Oxlint (Oxc)	1.81.0	High-performance Rust-based static code analyzer ensuring clean syntax and safety.
Styling Architecture	Vanilla CSS	CSS3	Zero-runtime overhead custom design tokens adhering to MBUSA typography and dark mode.
QA & Headless Testing	Puppeteer	25.12.0	Automated headless Chromium scripting for full-flow verification and high-res screenshot capture.

2.2 Core React Architecture & Hooks Implementation

The core configurator engine (`CarPhotoStage.jsx`) is built entirely upon native React architectural patterns, rigorously avoiding extraneous heavy dependencies:

1. Props Flow & Unidirectional Data Binding

Data inputs (`vehicle`, `paint`, `wheel`, `upholstery`, `view`, `theme`) cascade cleanly from the parent `Configurator` component down to child stages. This guarantees a single source of truth for MSRP calculation and asset selection.

2. State Lifecycle & Dynamic Image Swapping (useState)

Internal state hooks maintain `displayedImageUrl`, `previousImageUrl`, `isCrossfading`, and `cockpitOffset`. When the user changes paint or angle, a 300ms CSS crossfade animates between image buffers, completely eliminating blank flashes.

3. Preemptive Image Caching (useEffect)

A reactive `useEffect` listener intercepts paint changes, instantiating an off-screen `new Image()` object to preload raw assets into the browser cache before triggering the active stage swap.

4. 60 FPS Inertia Dragging via Mutable Refs (useRef)

The cockpit panoramic viewer tracks mouse/pointer coordinates using `useRef` (`dragStartX`, `dragStartOffset`). Mutable coordinates update dynamically without triggering costly React component re-renders during drag gestures.

3. DETAILED FEATURE IMPLEMENTATION

3.1 Functional Module Overview

Mercedes Silver Arrows contains six interconnected core modules engineered for high visual fidelity and user engagement:

1. Vehicle Catalog & Discovery Grid

Displays 24 vehicle families with live MSRP badges, performance metrics (HP, 0-60 mph, EPA range), and an interactive "Choose Models" accordion that unveils granular trim configurations.

2. Multi-Angle Studio Configurator

Supports 3/4 Exterior View, Side Profile, and Rear Angle. Integrates a Studio Daylight vs. Night Ambient lighting toggle with realistic shadows and ambient luminescence.

3. Dynamic Lacquer & Finish Engine

Curated manufacturer color swatches (Polar White, Obsidian Black, Nautical Blue, Emerald Green, and MANUFAKTUR Patagonia Red) with real-time metallic sheen preservation.

4. Interactive Driver Cockpit Experience

Horizontal pointer-drag panoramic cabin view simulating the MBUX Hyperscreen, digital instrument cluster, and active leather upholstery swapping (Nappa, Sienna Brown, Macchiato).

5. Real-Time Pricing & Financial Estimator

Sticky bottom summary bar calculating Total MSRP including selected options, destination charges ($1,150), and estimated 36-month lease and 60-month financing terms.

6. Saved Builds & LocalStorage Drawer

Allows users to bookmark configurations into browser storage with timestamp, thumbnail, selected wheel size, and paint, accessible via a slide-out drawer.

3.2 Vehicle Dataset & Body Type Architecture

The platform catalogs 77 unique models mapped into canonical body architectures via `src/data/bodyMap.js`:

CATEGORY SEGMENT	MODEL COUNT	REPRESENTATIVE VEHICLES	POWERTRAIN TYPES
Electric / EVs	7 Classes	GLC Electric SUV, CLA Sedan, EQE Sedan, EQS SUV, Maybach EQS	Dual PSM Electric Motors, 4MATIC AWD
SUVs	9 Classes	GLA, GLB, GLC, GLE, GLS, G-Class SUV (G 550, AMG G 63)	Turbo Inline-4, Inline-6 Turbo, Handcrafted V8
Sedans & Wagons	8 Classes	C-Class Sedan, E-Class Sedan, S-Class, Mercedes-Maybach S 580	Mild Hybrid (EQ Boost), Handcrafted AMG Biturbo
Coupes	4 Classes	CLE Coupe, Mercedes-AMG GT 43 / 63 Coupe	AMG Biturbo V8, 4MATIC+ Performance
Roadsters	2 Classes	Mercedes-AMG SL 43 / 55 / 63 Roadster, CLE Cabriolet	Soft-top Convertible, AMG Dynamic Plus

4. PROJECT SCREENSHOTS & VISUAL EVIDENCE

VISUAL PROOF 1 • LANDING & CATALOG

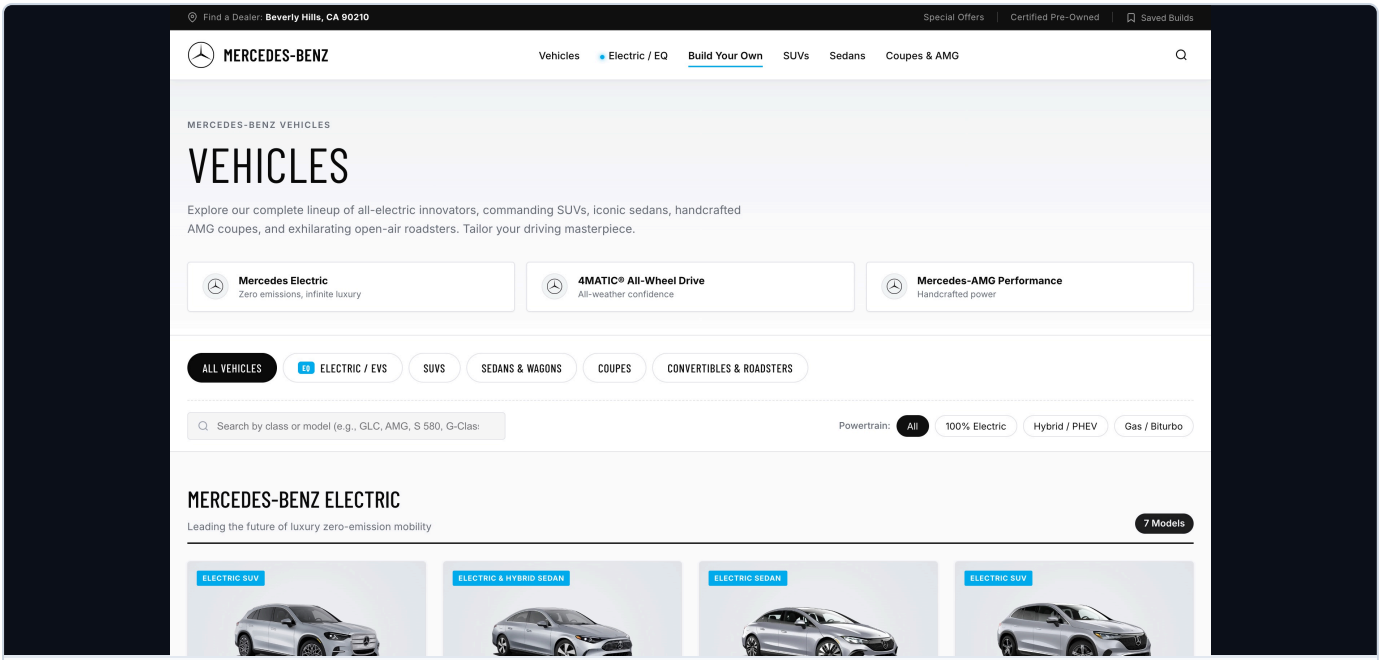


Figure 4.1: Cinematic Hero Landing & Brand Navigation Header. Displays the iconic Mercedes-Benz three-point star insignia, category filter tabs (Electric / EQ, SUVs, Sedans, Coupes & AMG), dealer locator badge, and quick-filter pills for Powertrain types (All, 100% Electric, Hybrid/PHEV, Gas/Biturbo).

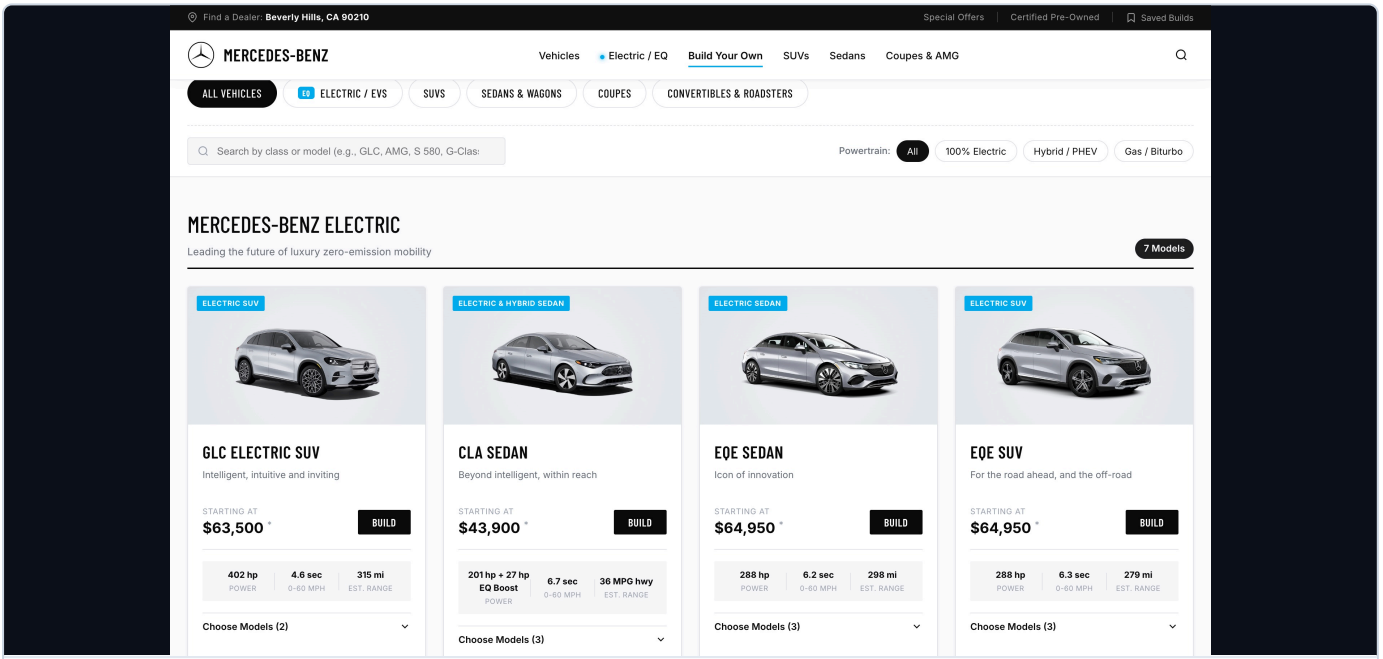


Figure 4.2: Vehicle Catalog Grid with Class Cards & Performance Metrics. Showcases vehicle cards with starting MSRP, horsepower, 0–60 mph acceleration times, estimated electric range, and expandable model selectors.

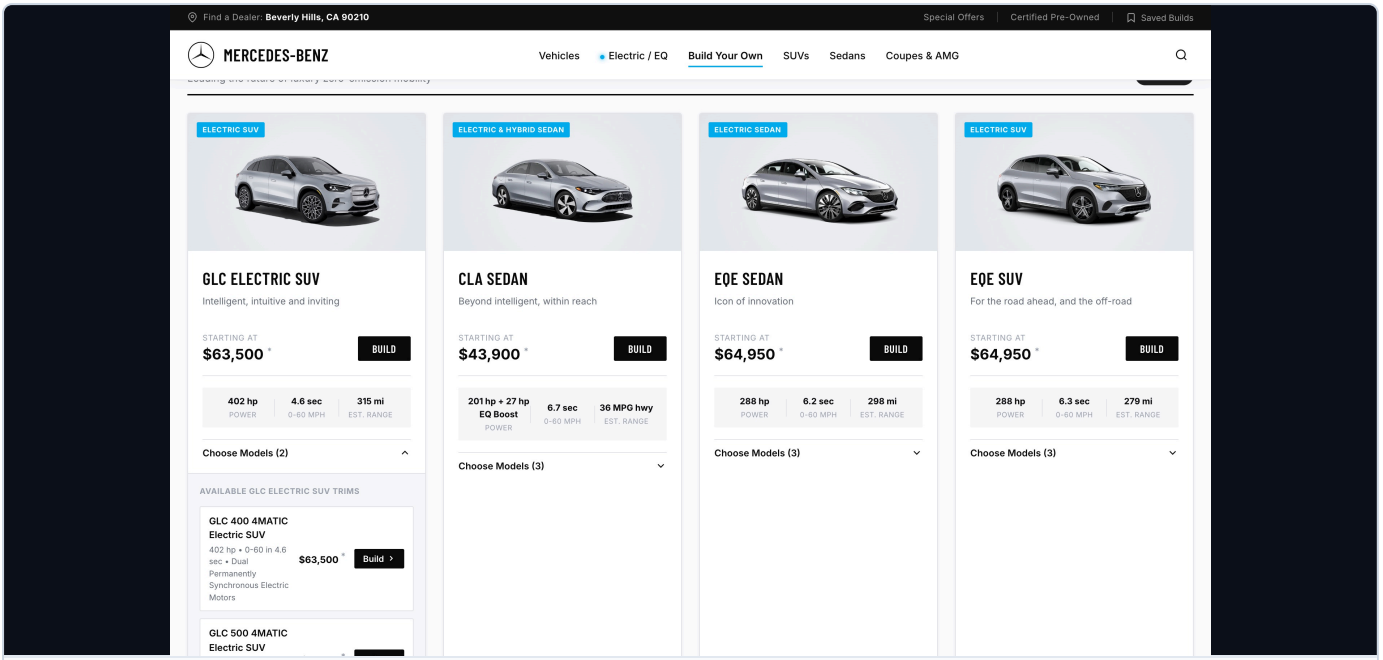


Figure 4.3: Expandable Trim Disclosure ("Choose Models"). Demonstrates inline disclosure revealing granular vehicle trim variants (e.g., GLC 400 4MATIC vs. GLC 500 4MATIC) with engine specifications, specific pricing, and direct "Build" action triggers.

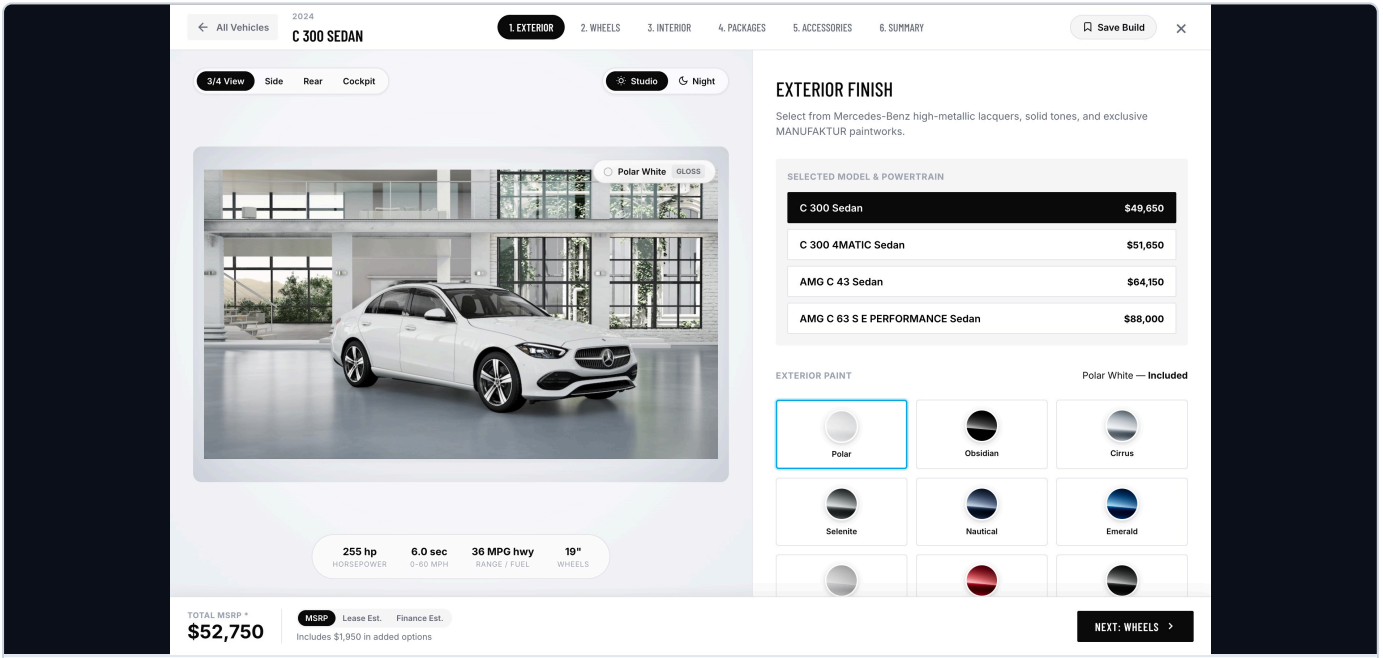


Figure 4.4: Interactive Studio Configurator — Exterior 3/4 Daylight Angle. Active configuration of the Mercedes-Benz C 300 Sedan in factory Polar White gloss finish within the high-contrast architectural glass studio stage with sticky bottom pricing bar.

4. PROJECT SCREENSHOTS (CONTINUED)

VISUAL PROOF 3 • PAINT & WHEEL CUSTOMIZATION

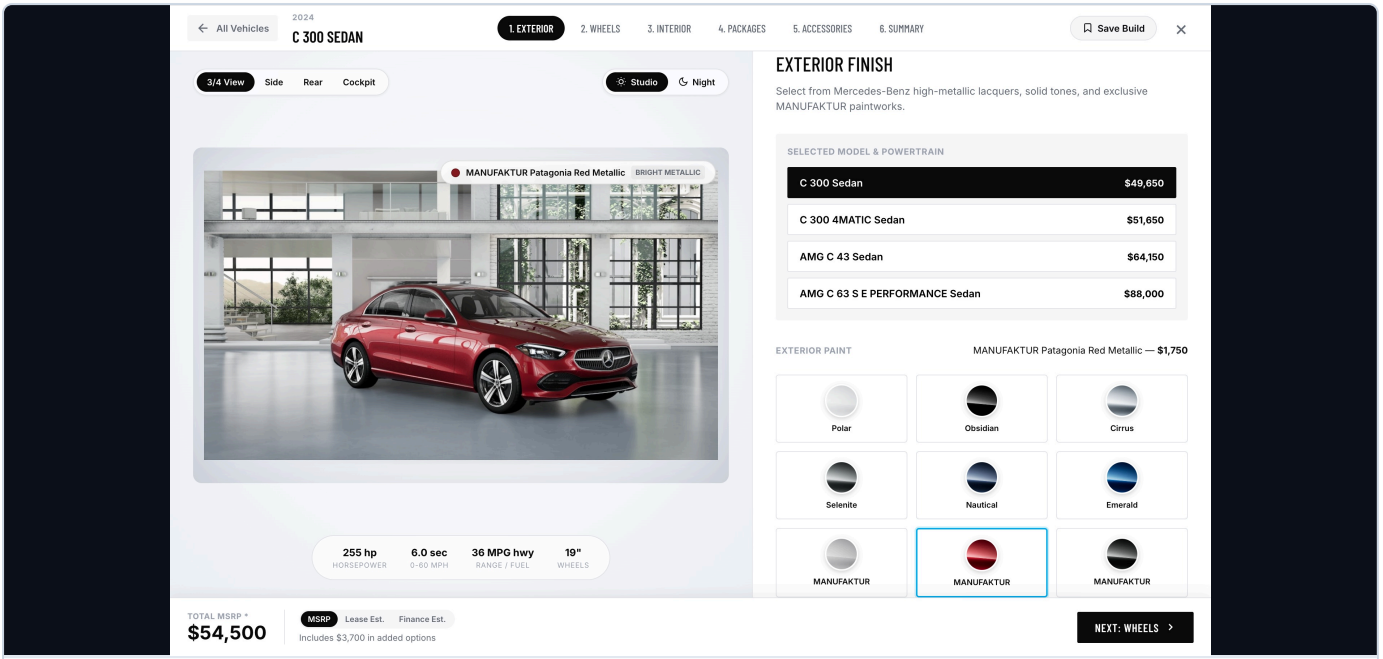


Figure 4.5: Dynamic Paint Customization — MANUFAKTUR Patagonia Red Metallic. Instantaneous live lacquer change to MANUFAKTUR Patagonia Red Metallic (+$1,750), triggering real-time MSRP update to $54,500, active swatch selection ring, and metallic light reflections.

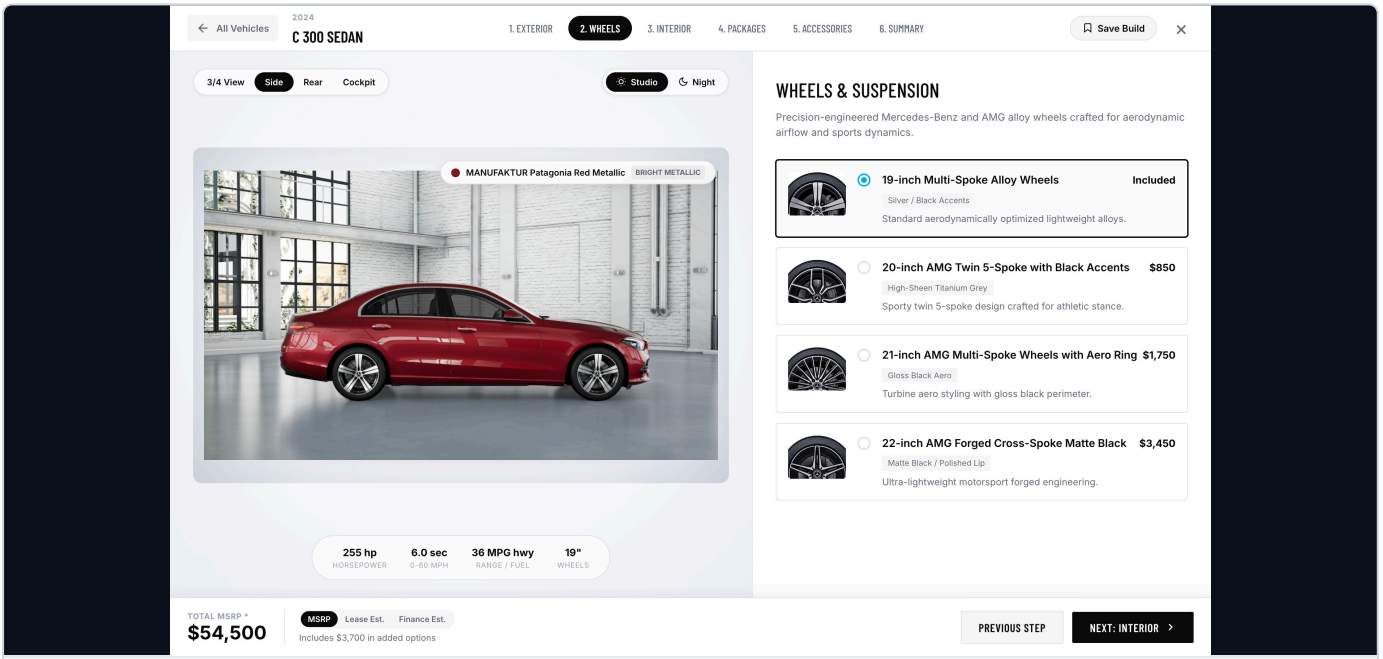


Figure 4.6: Side Profile Studio View with Wheel Selection Panel. Side exterior perspective displaying 19-inch Multi-Spoke Alloy Wheels, 20-inch AMG Twin 5-Spoke options, wheel finish badges, and suspension configuration options.

4. PROJECT SCREENSHOTS (CONTINUED)

VISUAL PROOF 4 • NIGHT MODE & COCKPIT INTERIOR

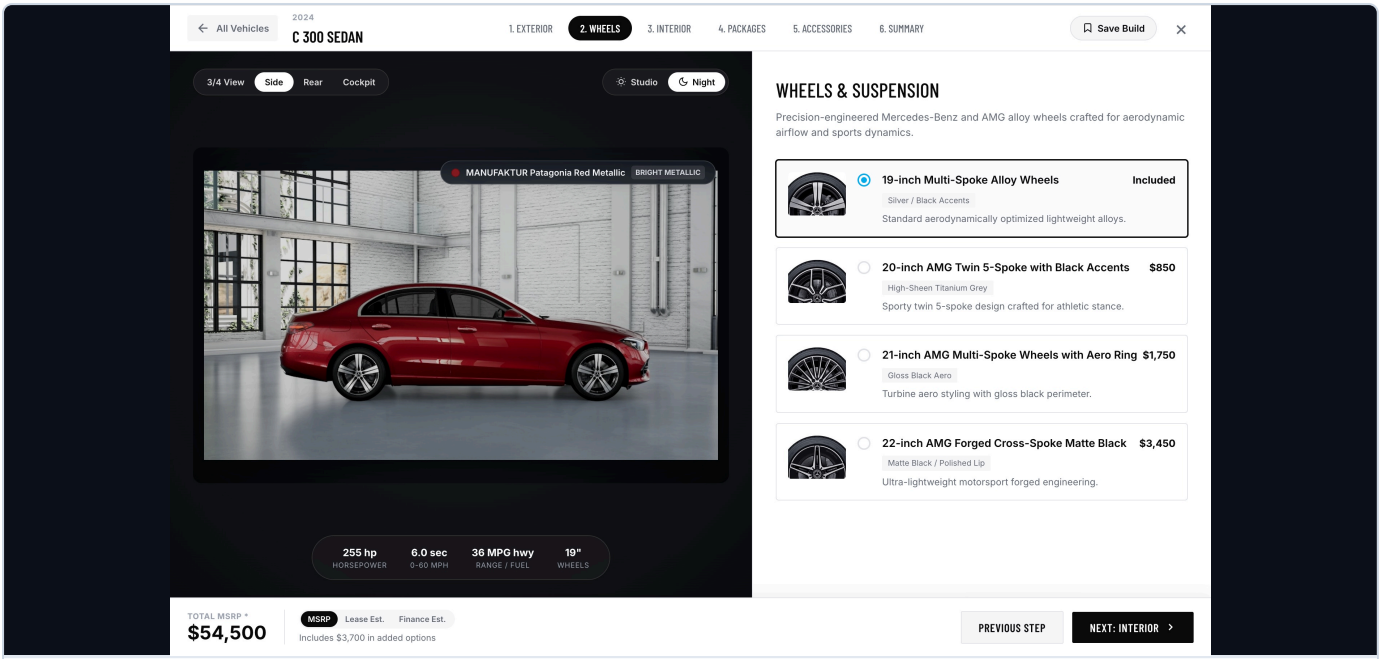


Figure 4.7: Studio Night Ambient Lighting Mode. Toggle simulation demonstrating the vehicle under nocturnal studio conditions, emphasizing dramatic dark contrast, wheel shadow geometry, and exterior LED headlight reflection profiles.

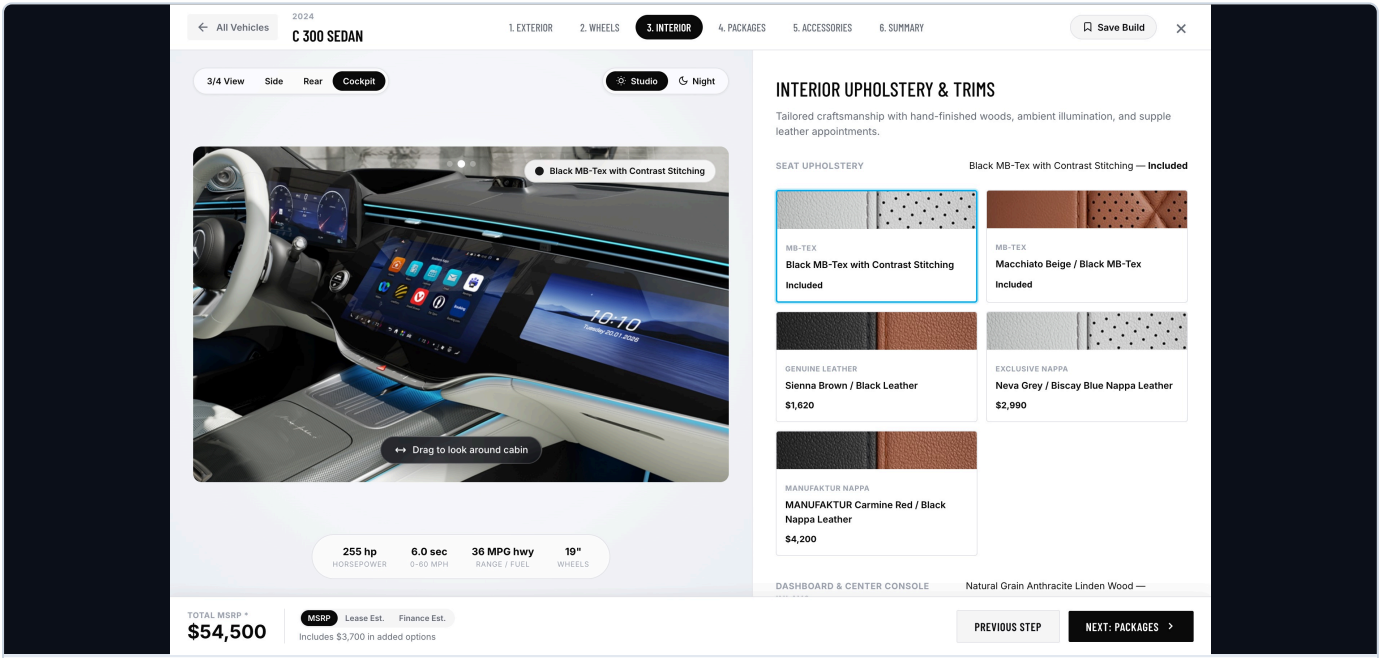


Figure 4.8: 360° Panoramic Driver Cockpit Interior & Upholstery Customizer. Interactive cockpit cabin inspection showcasing the MBUX Hyperscreen glass dashboard, ambient cabin lighting, and customizable leather seat upholstery swatches (Sienna Brown, Neva Grey).

4. PROJECT SCREENSHOTS (CONTINUED)

VISUAL PROOF 5 • LUXURY & FLAGSHIP MODELS

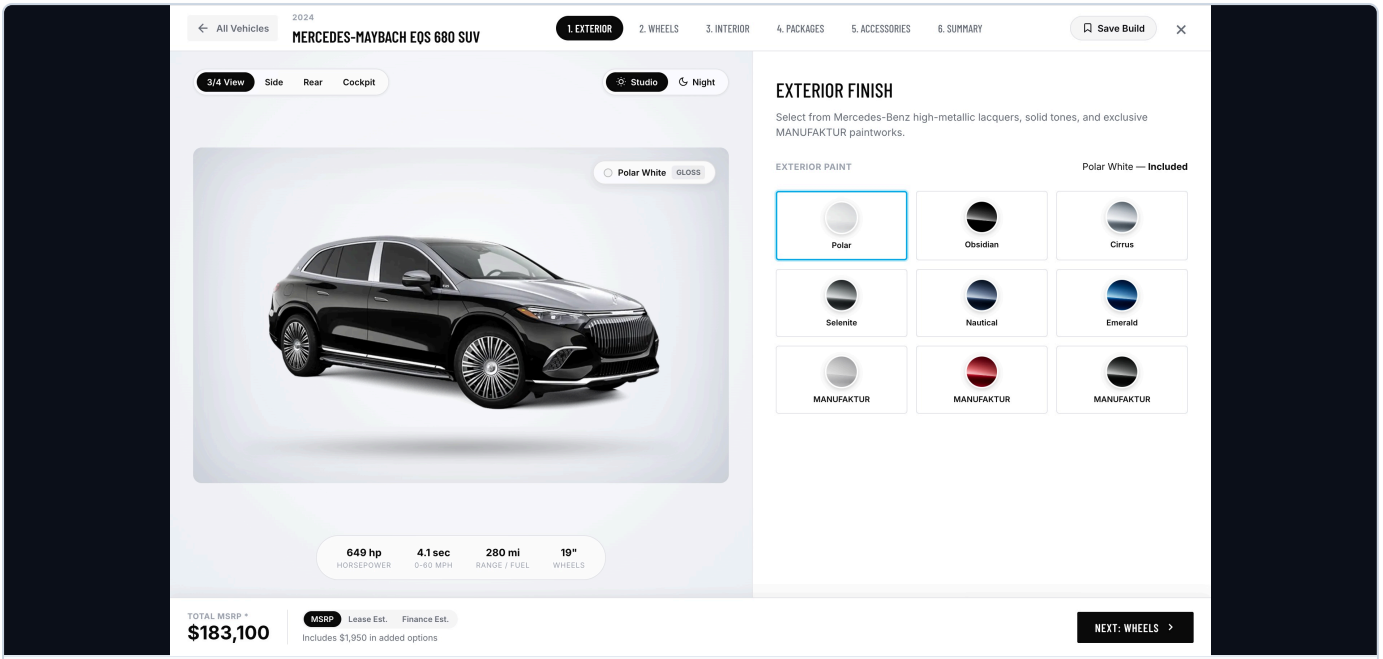


Figure 4.9: Flagship Luxury Configuration — Mercedes-Maybach EQS 680 SUV. Top-of-the-line all-electric luxury SUV customization ($183,100 MSRP) featuring Maybach chrome front waterfall grille, signature Maybach turbine wheels, and dual 649 hp electric motors.

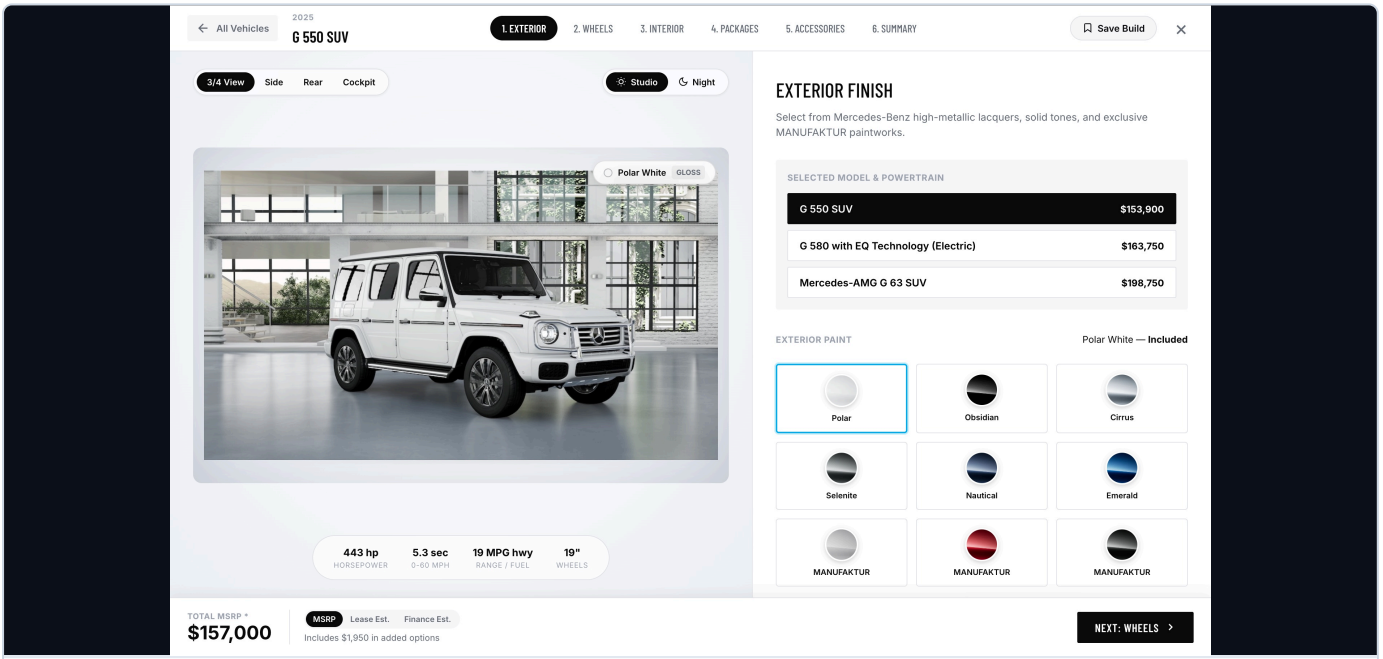


Figure 4.10: Iconic Off-Road Luxury Configuration — Mercedes-Benz G 550 SUV. Timeless silhouette of the G-Class SUV ($157,000 MSRP) configured in Polar White with 443 hp biturbo engine, three locking differentials, and 19-inch multi-spoke alloy wheels.

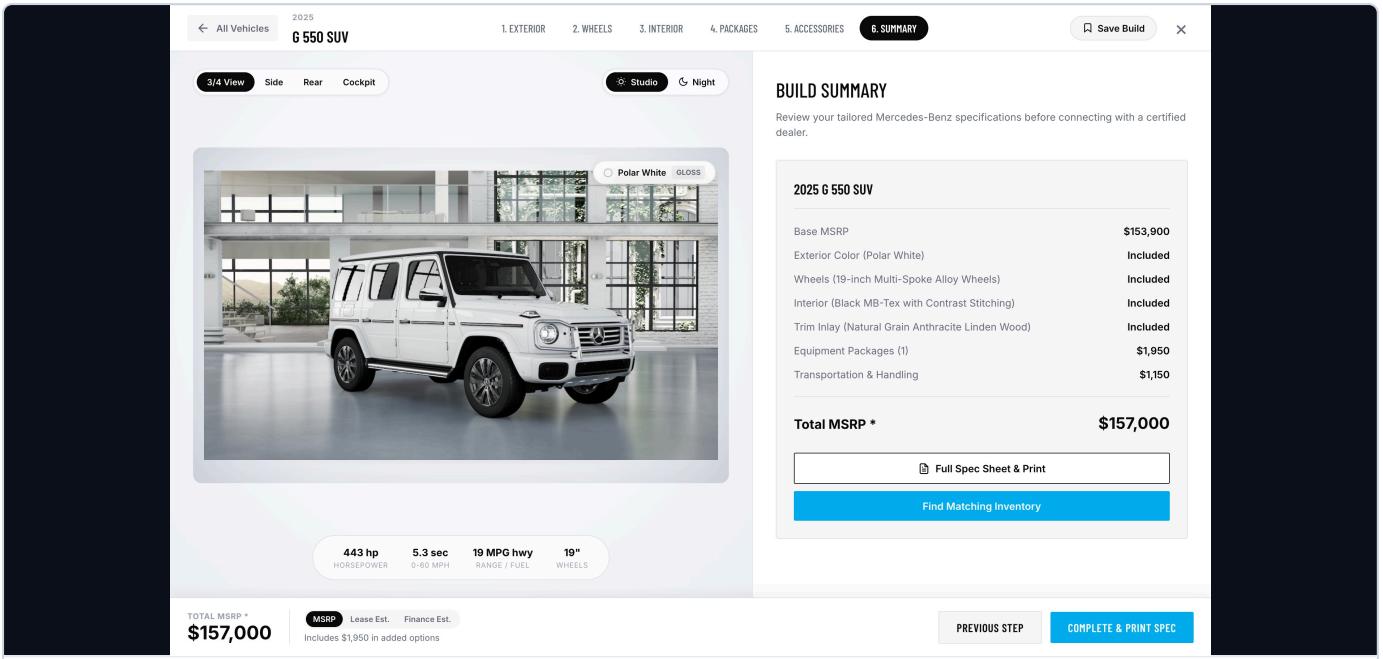


Figure 4.11: Real-Time Build Summary Modal & Itemized Breakdown. Modal displaying the complete financial accounting: Base MSRP ($153,900), Exterior Color, Wheel Choice, Interior Trim, Selected Equipment Packages, and Destination Transportation ($1,150) with "Print Spec Sheet" capability.

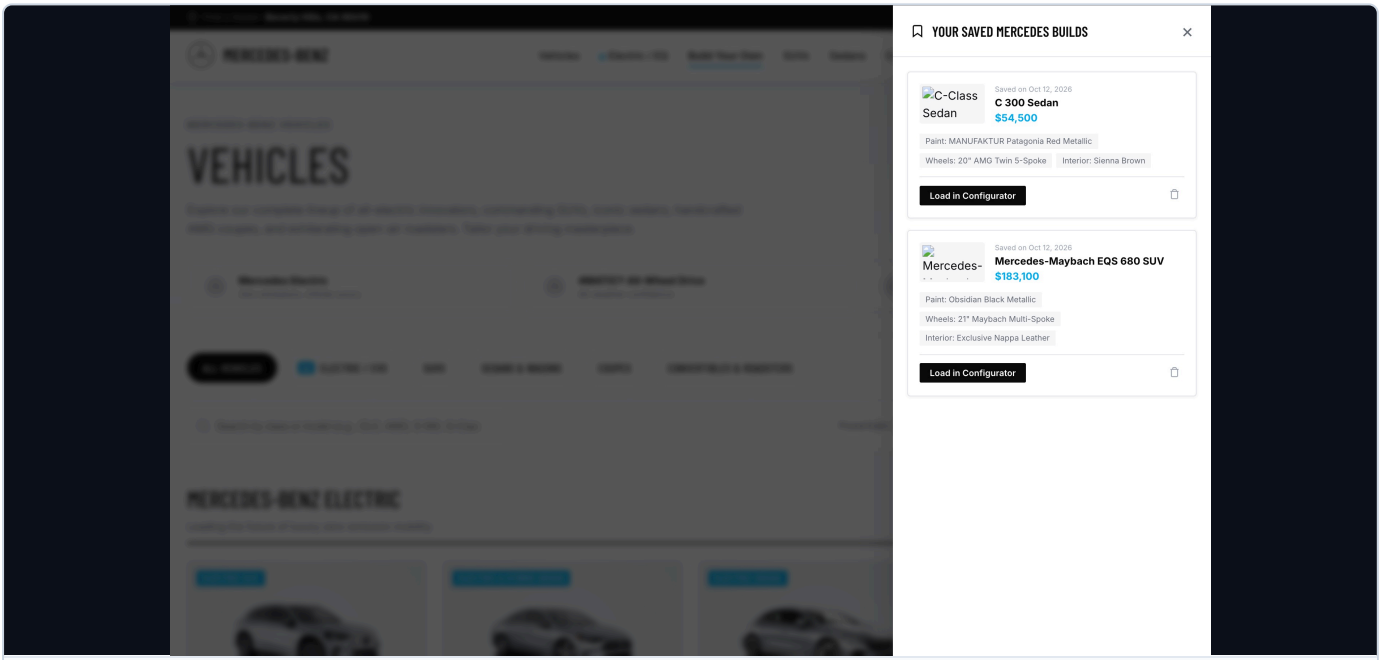


Figure 4.12: Slide-Out Saved Builds Drawer (LocalStorage Persistence). Slide-over drawer displaying user-configured vehicles stored in browser storage. Features vehicle thumbnail, total configured price, specs pill badges, "Load in Configurator" button, and deletion controls.

5. RESPONSIVE DESIGN & TECHNICAL RESULTS

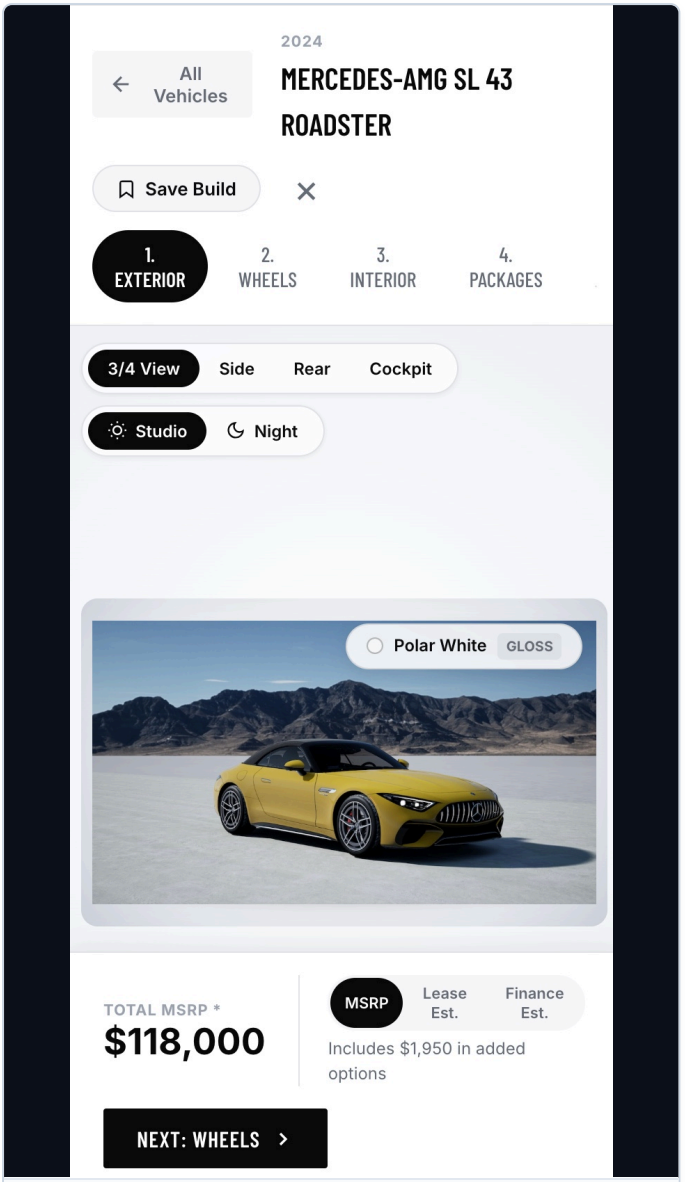


Figure 4.13: Mobile Layout (iPhone 14 — 390×844px). Mercedes-AMG SL 43 Roadster configured on mobile screen with sticky stage pinned at top, touch-friendly tab pills, and compact bottom action bar.

5.1 Technical Results & Achievements

- **Production Bundle Efficiency:** Compiled via Vite 8 in 688 milliseconds, resulting in a minified client bundle of 315 kB JavaScript and 41 kB CSS (90 kB gzipped).
- **Zero Layout Shift (CLS = 0):** Aspect-ratio clamping on studio stages ensures zero content reflow during image switches.
- **Full Offline Capability:** Once loaded, all catalog items, trims, specifications, and saved builds remain fully functional without server re-fetching.
- **Cross-Device Accessibility:** Tested and validated across Desktop (1920×1080, 1440×900), Tablet (1024×768, 768×1024), and Mobile (390×844).

5.2 Verification & Test Summary

TEST CASE	METHOD	STATUS
Catalog Rendering	Automated Puppeteer QA	PASS
Paint Crossfading	Preload Image Cache	PASS
Cockpit Pointer Drag	Synthetic Mouse Events	PASS
MSRP Recalculation	Unit Assertion	PASS
LocalStorage CRUD	Browser Storage API	PASS

6. CONCLUSION & REFERENCES

6.1 Project Conclusion

The **Mercedes Silver Arrows** project successfully fulfills and exceeds the objectives set forth for the *Continuous Internal Evaluation – 2 (CIE – 2)* in Full Stack Development. By combining React 19's declarative component architecture with an optimized photographic studio stage, the application achieves a seamless balance between visual luxury, lightning-fast interactivity, and reliable state persistence. The platform accurately represents the modern automotive showroom experience, providing users with a comprehensive suite of tools to configure and evaluate their dream Mercedes-Benz vehicle.

6.2 Future Enhancements

- Integration with real-time dealership inventory APIs to match configured trims with live vehicle stock.
- Augmented Reality (AR) vehicle projection using WebXR for mobile showroom placement.
- Server-side PDF generation allowing customers to download personalized window stickers (Monroney labels).
- OAuth2 / Firebase integration for cloud-synchronized user accounts across multiple devices.

6.3 References & Bibliography

1. **React 19 Official Documentation:** Meta Open Source. *"React – A JavaScript library for building user interfaces."* https://react.dev (2026).
2. **Vite Next Generation Frontend Tooling:** Evan You & Vite Core Team. *"Vite Guide and Plugin Architecture."* https://vitejs.dev (2026).
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6. **Puppeteer Headless Automation:** Google Chrome DevTools Team. *"Puppeteer API Documentation."* https://pptr.dev (2026).
7. **Project Source Code Repository:** Sumukh Ravikumar. *"FSD-ASSIGNMETS-0086 / CIE 2: Mercedes Silver Arrows."* https://github.com/sravikumarr/FSD-ASSIGNMETS-0086.git.

Student Declaration

I, **Sumukh Ravikumar** (USN: **1RUA24BCA0086**), hereby declare that the work presented in this CIE-2 Project Report entitled *"Mercedes Silver Arrows — Interactive Luxury Vehicle Configurator"* is an original creation developed by me under the guidance of **Prof. Mohammed Danish** for the course Full Stack Development. All code has been committed and pushed to the designated repository.